TLA$^+$: temp. logic of actions $\Box$, $\Diamond$ ← eventually

always

SMV: LTL + CTL

lin. time logic

computation tree logic

$\boxed{\text{LTL:}}$ $\varphi ::= \top \mid a \mid \varphi_1 \wedge \varphi_2 \mid \bigcirc \varphi \mid \varphi_1 \, U \, \varphi_2 \mid \neg \varphi$

$a \in AP$

next

φ_1 until φ_2

more connectives:

$\varphi_1 \vee \varphi_2 := \neg(\neg \varphi_1 \wedge \neg \varphi_2)$

$\varphi_1 \rightarrow \varphi_2 := \neg \varphi_1 \vee \varphi_2$

$\varphi_1 \leftrightarrow \varphi_2 := (\varphi_1 \rightarrow \varphi_2) \wedge (\varphi_2 \rightarrow \varphi_1)$

$\varphi_1 \oplus \varphi_2 := (\varphi_1 \wedge \neg \varphi_2) \vee (\neg \varphi_2 \wedge \varphi_1)$ XOR

temporal conn.: now t_0 t_0+1 t_0+2 ...

atomic a: $a \to$ arbitrary $\to$ arb. $\to$ arb. $\to \dots$

next $\bigcirc a$: arb. $\to a \to$ arb. $\to$ arb. $\to \dots$

until $a \mathcal{U} b$: $a \wedge \neg b \to a \wedge \neg b \to a \wedge \neg b \to b \to a \wedge b \to \dots$

eventually $\Diamond a$: $\neg a \to \neg a \to \neg a \to a \to$ arb. $\to \dots$

always $\Box a$: $a \to a \to a \to a \to a \to \dots$

$\Diamond \varphi := \top \mathcal{U} \varphi$ $\Box \varphi := \neg \Diamond \neg \varphi$

$\Box \Diamond \varphi$: "infinitely often φ"

$\Diamond \Box \varphi$: "eventually forever φ"

mutual exclusion:

race condition

banking in TCAT

P_1, P_2

1) noncrit.
2) waiting
3) crit. sec

safety:

$\Box(\neg critical_1 \vee \neg critical_2)$

liveness:

$(\Box\Diamond critical_1) \wedge (\Box\Diamond critical_2)$

Semantics (over (infinite) words):

φ LTL-formula over some set AP of atomic propositions.

$2^{AP} = \mathrm{Fun}(AP, 2)$
$= \mathrm{Fun}(AP, \{0,1\})$

Truth value assignments

LT-property induced by φ:

$$\mathrm{Words}(\varphi) := \{ \sigma \in (2^{AP})^\omega \mid \sigma \models \varphi \}$$

- σ is a sequence of valuations of AP

"σ satisfies φ"

In propositional or predicate logic, <u>satisfaction</u> is a relation $\models \subseteq 2^{AP} \times \text{Form}$

But in LTL satisfaction is a relation $\models \subseteq (2^{AP})^{\omega}$

$\cong$

$\text{Fun}(\mathbb{N}, \text{Fun}(AP, 2))$

$\models$ generated by:

$\sigma \models \top$

$\sigma \models a$ iff $A_0 \models a$

Valuations are sequences

$\sigma = A_0 A_1 A_2 \ldots$